

MTA NYCT Subway Fare Evasion

Overview

General Description

The Metropolitan Transportation Authority (MTA) is a public-benefit corporation responsible for public transportation in the state of New York serving 12 counties in southeastern New York, along with two counties in southwestern Connecticut under contract to the Connecticut Department of Transportation (CDOT). The MTA is the largest transportation network in North America.

Subway service within New York City is operated by MTA New York City Transit (NYCT).

Fare evasion occurs when passengers enter the subway system from the unpaid to the paid side by interacting with fare controls in manners inconsistent with the tariff—in other words, by not paying the fare. Understanding subway fare evasion is important for estimating subway ridership and gauging lost revenue. The data is used regularly by NYCT's Office of Management and Budget (OMB), and the Division of Operations Planning (OP).

The fare evasion rate is calculated systemwide and is estimated quarterly for the subway with a margin of error. The fare evasion rate is the estimated percentage of riders who illegally enter the system. It is estimated using surveys conducted by traffic checkers and by applying stratified sampling.

The fare evasion rate began being publicly reported in MTA Board meeting materials in April 2019.

This dataset was published during the first phase of the MTA's commitment to increasing transparency. We continually examine all our published and publishable data with a view to both providing datasets that can be effectively utilized by our customers and the public at large, and to providing regular, automated updates to these datasets efficiently and sustainably. Consequently, this dataset may be restructured and/or combined with other similar datasets in the future.

Data Collection Methodology

The fare evasion rate is estimated using a statistical survey, which is a study that obtains or records data from a subset of a population in order to estimate attributes of the entire population. A survey is used since it is unfeasible to have traffic checkers survey every fare control area in a station, and at all times of day. Though the sample only consists of about 0.03 percent of the 2.1 million combinations per

quarter of unique control areas (970) and unique hours (2,160), the statistical techniques are designed to ensure that the sample accurately estimates the fare evasion rate for the whole system.

Surveys only produce point estimates of the true population value, which is unknowable without doing an impossible 100 percent sample, and even then, the results would be subject to data collection errors. To address the statistical uncertainty of a sample estimate, a margin of error is calculated using a formula based on the type of sample, sample size, and variance within the survey result. The error margin is then used to provide a confidence interval around the point estimate – in this case tied to a 95 percent confidence level. This margin of error conveys how far away the surveyed result is likely to be from the true population value given the limited number of samples.

The surveys that traffic checkers take of fare evasion at selected fare control areas and at specific times are designed to be representative of the fare evasion rate for the entire system and at all times of day. While checkers will inevitably miss some fare evaders due to the limits of human observers, the fare evasion rate should be accurate.

The current method for measuring fare evasion in the subway was developed with assistance from Columbia University data analytics professor Dr. Kitty Kay Chan and the Office of the MTA Inspector General and was first used in Q1 2021, with a slight change made in Q4 2023. This method is not apples-to-apples comparable to the previous rates calculated using the previous methodology, though a dramatic change in the estimated rate due to the methodology change was not expected. Since this dataset includes data back to Q1 2018, before the change in methodology, this document will detail the methodologies for both approaches, note the differences between them, and changes made in 2019 to improve the previous approach.

The fare evasion rate is not reported at the station level given the very limited sample size, and since most fare control areas are not surveyed each quarter.

The current approach uses stratification and weighting since it is believed to produce more robust estimates of fare evasion that are less susceptible to the variability of simple random sampling.

The methodology for the current approach to calculating fare evasion is detailed below:

Sample Generation

Obtaining paid entry data

The first step is to obtain a list of paid entries at each fare control area (FCA), which is a grouping of turnstiles, High Entry Exit Turnstiles (HEETS), and service gates, in each hour of each day of the first month of the prior quarter for which the sample is to be generated, from ridership data. Before Q3 2023 this data was provided by the Office of Management and Budget (OMB). A station may have multiple FCAs. This data is associated with three day types (Weekday, Saturday, Sunday) to capture different ridership patterns on different days of the week. Since ridership is fairly consistent during the week, only averaging weekday data, rather than averaging specific days, allows more flexibility for scheduling without losing too much accuracy. Since this data will initially include only one record for an average weekday, the weekday records will be repeated four additional times for a final population of FCA-hour combinations. This will better reflect reality (1 Saturday/1 Sunday/5 Weekdays per week).

Excluding combinations

This list of combination of FCA-hour type combinations is then updated for permanent and long-term closures of FCAs. FCAs used for special events, and FCA-hour types with zero entries for the full first month of the previous quarter are excluded. In addition, combinations between 11 p.m. and 6 a.m. are excluded since these hours comprise a very small percentage of overall ridership, and to allow for sensible scheduling with our traffic checker department. We believe that little accuracy is sacrificed by excluding overnight hours from the sample. For Q1 and Q2 2021, combinations between 10 p.m. and 7 a.m. were excluded due to the 1 a.m. to 5 a.m. overnight subway closure and a travel buffer to allow checkers to get to and from assignments.

Stratification

Fare evasion is measured using stratification, which divides the population of FCA-hour combinations into subgroups (strata) according to characteristics deemed significant and/or for which separate results are desired, which is total paid ridership. A statistical requirement for stratification is that the members of each stratum be relatively homogenous. Previous surveys found that fare evasion rates and FCA entries during the sample period are inversely related. As such, the strata are based on ridership to group FCA-hour units with more similar fare evasion behavior together.

Since fare evasion is measured using a stratified sample, the FCA-hour type combinations are split among six strata based on Automated Fare Collection paid ridership totals from turnstiles, defined as consisting of equal proportions of total paid ridership. Ridership is highly variable across stations, and within a station-across time and FCAs. By stratifying in this way, more surveys are conducted within low-

ridership parts of the system that have higher variance in measured fare evasion rates, which serves to reduce the statistical uncertainty of the overall, system-wide fare evasion estimate.

Each FCA-hour type combination, which is a sampling unit (also known as a cluster), is assigned to one of the six strata based on its total paid entries using the data obtained from ridership data.

Determining the number of surveys

The total number of surveys to be completed in a quarter is based initially on typical quarterly completion rates and checker availability. Based on past averages, the expected number of completed surveys per quarter is 490. Assuming an 80 percent completion rate, the total number of surveys that will be chosen for the quarter is 612. Completed surveys are monitored to ensure they are free from observational and data entry error.

Determining the number of surveys in each stratum

The method used to determine the number of surveys in each stratum was revised in Q4 2023. Both methods are detailed here.

Method from Q4 2023 to present

The number of surveys assigned to each stratum is determined using this algebraic formula, which works regardless of how many surveys are assigned to each stratum, meaning it can handle disproportionate sampling):

$$Strata\ Weight = \frac{\frac{1}{(6 \times (1 - Strata\ FE))}}{\sum \frac{1}{(6 \times (1 - Strata\ FE))}}$$

A factor was calculated by dividing the targeted monthly surveys by the total number of eligible FCA-hour type combinations and multiplying the result by the number of months in the period. Then the number of FCA-hours types assigned to each stratum was multiplied by this factor to get the survey distribution.

For example, based on October 2020 paid entry data, the strata boundaries and distribution of surveys are:

Strata	Lower Bound	Upper Bound	# of FCA-Hour- Day Type Combos	# of Surveys
1	≥ 1	< 70	217,204	329
2	≥ 70	< 118	74,167	112
3	≥ 118	< 177	48,152	73
4	≥ 177	< 260	32,015	49
5	≥ 260	< 417	21,133	32
6	≥ 417	< 2000	10,991	17
Total			403,662	612

In this case, the factor is: $(204/403,662) * 3 = 0.001516$, where 204 is the monthly targeted number of surveys, 403,662 the number of eligible FCA-hours in October, and three is the number of months in the quarterly sample period.

Method between Q1 2021 and Q3 2023

Between Q1 2021 and Q3 2023, a different formula was used to assign the number of surveys to each stratum. The surveys were assigned proportionately to a stratum's total number of entries.

Each survey had the weight of:

$$\text{Strata Weight} = \frac{\text{Total Strata Entries (from surveys)}}{\sum \text{Total Strata Entries (from surveys)}}$$

Survey assignment was updated slightly in Q4 2023 to use disproportionate sampling. This is a type of stratified sampling in which the size of the sample from each stratum is not in proportion to the size of that stratum in the total population. This method allows an arbitrary number of surveys to be assigned to any strata while still generating a statistically valid estimate of fare evasion.

The use of disproportional sampling was made necessary by the Agency's need to better understand fare evasion at busier stations without requiring additional surveys. Using disproportional sampling, busier stations could be more heavily sampled in a manner that further improves confidence in the system-wide Fare Evasion estimate, and weekend samples could be reduced to accommodate staff limitations without interrupting the statistical validity of the system-wide estimate.

Randomly choosing a sample

Each quarter, and for each of the six strata defined in the previous step, a sample is randomly pulled from the up-to-date population of FCA-hour type combinations, with a total number of surveys chosen corresponding to the number of FCA-hour types assigned to each stratum in the previous step.

Scheduling tasks

Schedule surveys

Operations Planning (OP) schedules surveys to match the samples, given available traffic checker resources. Each sample is completed only once per quarter. If the same FCA-hour-weekday combination appears multiple times in a sample, it should be done on a different day.

Reschedule surveys

OP staff regularly review which surveys were not completed so they can be rescheduled for another chance at completion during the quarter. They check that the surveys that are completed are reassigned to the same stratum, day, and FCA.

Completion of surveys

A group of ten traffic checkers goes to the appropriate FCAs at the assigned times and days and complete the surveys. They use an iPad app, or paper if there are technical issues with the device, to note how many individuals illegally entered through the fare gates over the course of an hour. Traffic checkers record every entry type they notice. The list of types of entries is legal (paid, authorized, school groups or children under 44 inches), illegal (High Entry/Exit Turnstile bump, swipe seller or open the gate) and questionable entries. Questionable evasion is when it is unclear if the person paid or needed to pay. including people entering with a gate key or wearing an official uniform, such as a police officer or person wearing an MTA vest. Checkers should survey from unpaid area unless their view is obstructed.

Evasion calculation

This section provides details on how the systemwide fare evasion estimate has been calculated since Q1 2021. For details on how fare evasion was calculated between 2018 and Q4 2020, look at the section on the previous methodology.

Survey results assigned to strata

This first step stopped being done after Q3 2023, to allow for a greater number of surveys to be measured at busy station complexes (also known as “disproportionate sampling”). Results were assigned to strata based on actual paid and illegal entries for each FCA survey hour, including FCA-hour combinations that reported no paid ridership. Since actual ridership for specific FCA-hour combinations often differed from the historic ridership for these combinations obtained from OMB, some surveys were reassigned to new strata. This process created six strata that were almost equally weighted by total entries (paid and unpaid ridership) with no overlap.

Calculating the evasion rate

The evasion rate and margin of error is then calculated within each stratum. The sampling unit for the calculations is an individual survey done by a traffic checker, which is also known as a cluster, which is a specific FCA-hour combination, and the elementary unit is a person entering the system.

The evasion rate within an individual stratum is calculated as the summation of the number of selected persons with the attribute of interest (illegal entries) in a specific cluster divided by the number of total selected persons or entries (legal + questionable + illegal) for that specific cluster. These calculations are done using the following formulas:

$$p_s = \sum_{i=1}^N \frac{a_{si}}{m_{si}}$$

$$m_{si} = L_{si} + Q_{si} + a_{si}$$

The margin of error for a specific stratum is calculated with a 95 percent confidence level using the formula for the confidence interval of a proportion, which is the z-score for the 95 percent confidence level (1.96) multiplied by the square root of the variance of the proportion of illegal entries to total entries across clusters in the stratum, or the fare evasion rate, $(v(p_s))$. The numerator for the variance of the proportion is the summation of the number of selected persons with the attribute of interest (illegal entries) in a specific cluster subtracted by the evasion rate for that stratum multiplied by the number of total selected persons or entries (legal + questionable + illegal) for the specific cluster to the second power. The denominator for the variance of the proportion is the number of sampled clusters multiplied by the number of sampled clusters minus 1 multiplied by the average number of selected persons (entries) per cluster to the second power. These calculations are done using the following formula:

$$ME_s = 1.96\sqrt{v(p_s)} = 1.96\sqrt{\frac{\sum_{i=1}^N (a_{si} - p_s m_{si})^2}{n_s(n_s - 1) \bar{m}_s^2}}$$

Where:

a_{si} is the number of illegal entries in sampled cluster i , stratum s

m_{si} is the number of entries in sampled cluster i , stratum s

L_{si} is the number of legal entries in sampled cluster i , stratum s

Q_{si} is the number of questionable entries in sampled cluster i , stratum s

$v(p_s)$ is the variance of p_s

p_s is the proportion of total number of illegal entries to total entries across sampled clusters in stratum s

n_s is the number of sampled clusters in stratum s

$\bar{m}_s$ is the average number of entries per sampled cluster in stratum s

Weighing results for each stratum

The systemwide evasion rate and margin of error are then obtained by averaging the stratum-level averages calculated in the previous step, with each stratum weighted equally by its estimated proportion of total ridership. In this case, total ridership includes both legal, questionable, and illegal entries, which is obtained by scaling up stratum-level paid entries by the strata's estimated evasion rate for the quarter.

The number of illegal and questionable entries for each stratum are totaled and divided by the total number of entries in that stratum to obtain the non-payment rate for that stratum. The total number of legal entries (swipes) at all the FCA-hour combinations in that stratum (not just those randomly selected to be surveyed) is divided by 1 minus the non-payment rate for that stratum to obtain the total number of estimated entries (legal + questionable + illegal) for each stratum.

Next, this total number of entries for each specific stratum is divided by the sum of total strata entries for all six strata, which will be the weight for each stratum. The systemwide evasion rate is obtained by summing up the weight for each stratum multiplied by the fare evasion rate for each stratum. This equation is seen here.

$$p = \sum_{s=1}^S W_s p_s$$

The systemwide margin of error is calculated with a 95 percent confidence level using the formula for the confidence interval of a proportion but modified to weight the margins of error for the six strata. This formula is the z-score for the 95 percent confidence level (1.96) multiplied by the square root of the summation of the share of entries in the system that are in a specific stratum multiplied by the variance of the proportion of illegal entries to total entries across clusters in the stratum, or the fare evasion rate, ($v(p_s)$).

These calculations are done using this formula:

$$ME = 1.96 \sqrt{\sum_{s=1}^S W_s^2 v(p_s)}$$

Where: W_s is the share of entries in the system that are in strata s

Previous methodology

This section details the methodology used to calculate the fare evasion rate between 2009 and Q4 2020, and briefly touches on the approach used before 2009. In Q4 2018, changes were made to how the margin of error was calculated, and in July 2019, several modifications were made to the survey process.

Before 2009

Before Q1 2021, a different method was used to calculate systemwide fare evasion. This method, first used in 2009, replaced the measurement of fare evasion by station agents, who counted evaders for one day a month to calculate a systemwide evasion rate. Station agent counts were factored up by 35 percent to account from undercounting expected by this methodology. The data were not independently verifiable. Since station agents had other duties while being in the token booth (selling MetroCards, providing rider information), distracted clerks could easily miss a few fare evaders. At the same time, with decreasing agent positions, fewer control areas were covered. A 2008 pilot study that measured fare evasion at randomly selected fare control areas found that the overall fare evasion rate was six times the station agent rate-2.1 percent, as opposed to 0.32 percent. In 2008, a new methodology was developed to calculate fare evasion using stratification parameters determined by

two pilots from that year that found that passenger entry rates and adjacent neighborhood income levels correlated most strongly with fare evasion observed. This method began to be formally used in 2009.

2009 to Q4 2020

The sample for fare evasion was based on a random sample of FCA-hour combinations, stratified by average hourly ridership. OMB sent OP a file with the average ridership based on a year of data for every hour of the day at each FCA, separated into weekday, Saturday, and Sunday categories. The combinations were sorted by day type and the number of swipes in descending order. Any rows with fewer than 10 entries in an hour and exit-only control areas were removed, as were FCAs used for events.

The total number of rows was then divided by the number of strata to find the number of FCA-hour combinations in each stratum. 40 strata were used, which corresponded to 40 buckets. Initially, there were two median income brackets by 20 ridership strata. The FCA-hour combinations with highest ridership would be in the first bucket, with the next ones in the following bucket, and so on. With each row having a bucket number, the order of rows inside a bucket was randomized. Each quarter, a new sample was pulled from the buckets, starting with the first unused row. The number of rows pulled from each bucket, or strata, is 5 for weekdays, 1 for Saturday, and 1 for Sunday, which added up to 280 surveys in each sample set for a given quarter (40 each for Saturday and Sunday, and 200 for weekdays), with equal representation from each of the buckets. This was designed to ensure there was a mix of high and low ridership FCAs in each quarterly sample. Weekday type samples are drawn five times more than Saturday or Sunday to keep the sample balanced.

After the sample set for a given quarter was generated, during the first week weeks of the last month of a quarter, an analyst in OP looked through and removed any long-term closures from the sample. Surveys were assigned to checkers based on their work schedule. Like the current process, checkers measured evasion using categories using tablets, or paper as a backup. However, they did so for 30 minutes within a 1-hour period they were assigned. A checker was expected to do three checks a day, spread across three boroughs when feasible.

Since checkers observed clusters of riders at selected FCAs in selected hours, the evasion rate was calculated as a cluster sample proportion. The sampling unit was a FCA-hour combination, and the elementary unit was a person entering the system. Like the current methodology, the evasion rate is

calculated based on evasion as observed by traffic checkers and paid ridership data during the same time period.

This is the sample proportion used to calculate the fare evasion rate. In the equation, p is the proportion of riders evading the fare, n is the number of clusters (surveys [4Q 2018 to 4Q 2020] or buckets [2009 to 3Q 2018]), m_i is the number of entries the checker observes in each cluster for 4Q 2018 to 4Q 2020 or the total entries observed by checkers for FCA-hours in each bucket for 2009 to 3Q 2018, i, a_{ij} is the evader status of person i, j (0 or 1) and m_{ij} is the existence status of person i, j (0 or 1). This is an unsimplified version of the same equation used to obtain the fare evasion rate for each stratum under the current method. The difference is that this equation is only calculated once-for all the data in all the strata-there are not separate calculations for each stratum which are then averaged using weights.

$$p = \frac{\sum_{i=1}^n \sum_{j=1}^{m_i} a_{ij}}{\sum_{i=1}^n \sum_{j=1}^{m_i} m_{ij}},$$

Margin of Error: 4Q 2018 to 4Q 2020

This margin of error formula multiplied the z-score for the 95 percent confidence level (1.96) by the square root of the variance of the proportion of illegal entries to total entries across sampled clusters, or the fare evasion rate, $v(p)$. This formula is almost completely identical to the one currently used, except for the fact that it is done for all data and is not done for each stratum and then averaged using weights.

$$ME = 1.96\sqrt{v(p)} = 1.96\sqrt{\frac{\sum_{i=1}^n (a_i - pm_i)^2}{n(n-1)\bar{m}^2}}$$

Where:

a_i is the number of illegal entries in sampled cluster i

m_i is the number of entries in sampled cluster i

$v(p)$ is the variance of p

p is the proportion of total number of illegal entries to total entries across sampled clusters

n is the number of sampled clusters

$\bar{m}$ is the average number of entries per sampled cluster

Margin of Error: 2009 to 3Q 2018

Before Q4 2018, a different equation was used to calculate the margin of error at a 95 percent confidence level. With this formula, the buckets were the clusters, not the individual surveys for a specific FCA-hour combination. The numerator for the variance in the formula is the summation of the proportion of persons with the attribute of interest (illegal entries) in the bucket (the fare evasion rate) subtracted by the fare evasion rate for all buckets to the second power. The denominator was the number of buckets multiplied by the number of buckets to the second power. These calculations were done using the following formula:

$$ME = 1.96\sqrt{v(p)} = 1.96\sqrt{\frac{\sum_{i=1}^n (p_i - p)^2}{n(n-1)}}$$

Where:

$v(p)$ is the variance of p

p is the proportion of total number of illegal entries to total entries across all buckets (the fare evasion rate for all buckets)

p_i is the proportion of total number of illegal entries to total entries across bucket i (the fare evasion rate for the bucket)

n is the number of buckets

Descriptions of methodology changes

At some point, in an effort to improve accuracy, OP began treating the sampling unit of a FCA-hour as a cluster for statistical purposes since it was thought that it would provide a more cost-effective and accurate estimate of fare evasion.

As part of the changes made between Q4 2018 and Q1 2021, areas with fewer than ten swipes per hour were no longer excluded from being sampled. In addition, the hours that FCAs were surveyed on weekends were adjusted from 6 a.m. to 11 p.m. to being 5 a.m. to 11 p.m..

Margin of error change in Q4 2018

Several years later, OP revised the margin of error calculation for fare evasion as of Q4 2018 from being a simple random sample to being treated as a one-stage clustered random sample. Cluster samples

divide a population into groups, and selects the groups at random, and gathers data from every subject within that cluster, unlike stratified random samples, which randomly chooses some subjects from every group. Each survey, or FCA-hour combination, would be treated as a cluster, instead of buckets, which would be ignored, for the MOE calculation. This change widened the margin of error.

The change was made since OP had begun to consider that it would be more appropriate to assume that the proportion of evaders is clustered rather than uniform, an assumption of the binomial distribution, which had been used to represent fare evasion in the methodology. Binomial distributions are used to estimate the probable outcomes of a series of events when each event has two possible outcomes. They are based on assumptions that are questionable when applied to fare evasion, such as the independence of each event, since it is likely that passenger decisions are influenced by the behaviors of other passengers. The initial methodology had assumed observations to be independent, even though that was not strictly true. In addition, the sampling plan had selected every cluster in the population, which is what cluster sampling seeks to avoid, and the formula that had been used also did not account for the fact there were different numbers of people in each bucket.

Process changes in July 2019

In July 2019, a series of changes was made to address concerns about the process for obtaining the data raised during the methodology review with Columbia University and the MTA Office of the Inspector General. These changes took effect with the Q3 2019 data and could explain part of the increase in reported subway fare evasion-with fewer fare evaders missed.

The MTAIG had found that only 60 percent of surveys for the six quarters through the end of 2018 were completed, and wondered what the impact of the shortfall of completion surveys was on the accuracy of the fare evasion calculation. To increase the number of surveys completed, several changes were made. Traffic checkers previously had limited availability. Subway traffic checkers were part of the Passenger Environment Survey (PES) checker group and spent most of their time there. Traffic checkers often could not be substituted for in their absence. A position could not be filled until the following pick if a checker left the job or was out-long term; this had a direct impact on the number of completed checks. Bus and subway checkers were combined into one dedicated and coordinated unit, increasing the number of available checkers, and allowing them to be moved around between modes depending on sampling needs. Subway checkers would be taken out of uniform; they previously had been in uniform while doing surveys since they also had other responsibilities in the PES group.

Schedulers would begin to assign checkers to multiple nearby locations, instead of being spread out, reducing the proportion of the time that checkers needed to travel between observation locations. The requirement that checkers visit three boroughs per day would be relaxed to only require them to visit two boroughs a day to improve efficiency. These changes enabled a doubling of observation time from 30 to 60 minutes. This reduced the needed number of checks to obtain the same level of precision, and increased the number of checks performed each month with resources reallocated.

Categories on the surveys were simplified, eliminating unnecessary distinctions, such as whether evasion through a gate was deliberate or opportunistic. This would allow checkers to concentrate more on quantifying evasion instead of trying to discern a particular evader's behavior.

Previously, at FCAs with multiple exits, checkers were instructed to observe all gates within their assigned FCA. However, observing more than one at a time could be challenging at times, meaning they could miss many evaders. An expanded list of fare control areas that split out over a hundred additional control areas was created. It accounted for recent construction, and feedback from checkers and their supervisors. The split-up control areas allowed checkers to monitor smaller areas, and thus be better positioned to get better sightlines at stations to collect more accurate data. Finally, checkers were trained or retrained on standard practices, the number of supervisors assigned to evasion was increased from 1 to 3 (it increased to 4 in September 2019), and automated reports were created to flag results for quality assurance.

Q1 2021 methodology changes

The Q1 2021 methodology changes were developed with assistance from Columbia University data analytics professor Dr. Kitty Kay Chan and the Office of the MTA Inspector General. The change reduced the number of strata from 40 to 6. FCA-hour combinations would no longer be divided equally among the strata; instead, they would be divided so each strata had equal levels of ridership, meaning there would be different numbers of combinations in each stratum.

While the previous approach used strata to obtain a more accurate sample by ensuring there were enough surveys in areas with different levels of ridership by organizing FCA-hour combinations by ridership, dividing them equally into 40 strata, and randomly choosing some combinations, they did not weigh and average the fare evasion rates of the various strata. With the Q1 2021 changes, the fare evasion rate and margin of error would be calculated for each stratum and would then be averaged using weights. This change meant that the methodology would no longer consist of a hybrid method,

which combined elements of different sampling techniques. FCA-hours would be kept as clusters for the calculation of the margin of error. Another change is that the ridership data used to assign FCA-hours to strata for survey assignment would now be from each hour of each day of the first month of the first month of the prior quarter as opposed to being from the previous year of data.

Statistical and Analytic Issues

Surveys conducted by traffic checkers using their best judgement to count and classify entries, therefore human error may take place. Data is available starting in 2018. Fare evasion was not measured for Q2 2020 due to the COVID-19 pandemic.

Limitations of Data Use

There are no limitations on the data at this time.

Release Notes

Version 1.0.0 release notes instituted with documentation updates (03/13/2025)